Mikhail Balkhaev

ML Engineer / Generative AI Engineer / Product Developer

Moscow, Russia m.balkhaev@gmail.com +7 966 006-60-68 t.me/balkhaev balkhaev.com



ML engineer working on generative AI: image and video diffusion, speech, LLM agents and the GPU infrastructure that runs them. I build these systems from the model pipeline and its evaluation to the product backend. Before 2024 I spent 13 years in web development, including VK (Delivery Club) and Yandex (Yandex Eats).

Experience

AIGF, luv.club, adorely.co, sinly.io — Mar 2026 – present
Generative AI Engineer

- AI companion platforms (18+) with image, video and 3D generation, chat and voice: a multi-tenant Hono backend, Redis/BullMQ workers and RunPod GPU jobs with idempotent submission.
- Identity-consistent photos with Qwen Image 2.1: an edit-consistency LoRA, up to four reference images and an automatic redraw when the face drifts.
- Video: text- and image-to-video with two-pass sampling and RIFE interpolation to 48 fps; face and full-character swap with MiniMax H3 and swap LoRAs.
- 3D stage to video: the reference render conditions only the first 5 of 20 sampling steps, which keeps the camera and composition while the model produces natural motion.
- Companion memory: LLM fact extraction with corrections, hybrid retrieval over OpenAI embeddings in pgvector, lexical reranking and LLM relevance scoring.
- Age-safety checks for generated media: face detection, a ViT age classifier and a logistic probe on CLIP ViT-B/16 embeddings trained on 9,574 labelled person crops. On held-out data, correct refusals rose from 283 to 350 of 400 and false refusals fell from 26 to 3 of 302.
- Tool-using Grok agents: a studio director that sets up shoots and reviews results, and a 3D scene editor with validated, revision-checked edits; REST and MCP API with OAuth 2.1.
- Voice: self-hosted Kokoro-82M TTS on CPU (ONNX, 54 voices) and Whisper recognition.

Funcorp, Cyprus — Nov 2024 – Mar 2026
Senior Product Developer

- Started on the main product, then built new AI products from scratch.
- Diffusion pipelines for images and video (Wan 2.2, LTX, SD, SDXL) in ComfyUI and DiffSynth-Studio; dataset collection and LoRA training.
- Text-to-speech: voice design and fine-tuning the model for natural-sounding speech.
- LLM features with Qwen3.5-9B, ChatGPT and Grok; RAG on Qdrant.
- Data and backend: Kafka, ClickHouse, PostgreSQL, Redis, Next.js, Hono, FFmpeg.

ML projects

Studio, film.balkhaev.com

AI film production: a Claude producer agent turns an idea into a script and staged shots, renders reference passes in Blender over MCP and generates the shots with H3. A take is accepted only after frame decoding, a vision-model review and a speech-transcription check.

MediaPipes, mediapipes.balkhaev.com

Image, video and speech generation on ComfyUI and vLLM: Qwen Image 2.1 INT8 with up to ten references, tiled upscaling to 8 MP, Higgs TTS 3 (4B). INT8 attention for H3 cut sampling from 13.0 to 6.8 s/step (a swap job from 129 to 80 s) and shipped after identity, lip-sync and LPIPS metrics and a blind review.

Inference

GPU scheduler for a home RTX 5080 shared by ComfyUI, local LLMs and Blender: admission by measured RAM and VRAM peaks, GPU sharing only after three comparable solo runs, warm-model residency and budgets that grow after out-of-memory errors.

Local LLM

Qwen3.8-Flash-Next (125B GGUF) on one RTX 5080 through Strata, with experts spread across VRAM, RAM and SSD, a 262k context and about 100 tokens/s; an OpenAI- and Anthropic-compatible gateway with multi-account routing.

Montage
Video understanding: Qwen3-VL-4B frame descriptions, word-aligned WhisperX transcripts and PySceneDetect cuts in one timeline, with models loaded in turn to fit one GPU; video-to-video recasting with H3.

Modeler, modeler.balkhaev.com
Characters and previz in Blender: image-to-3D with TRELLIS.2 and Pixal3D to textured GLB, text-to-motion on a 63-bone rig with foot-contact quality gates, Qwen-guided facial reconstruction.

Web development

Yandex, Moscow — Oct 2022 – Aug 2024
Frontend Developer

- Moved from VK with the Delivery Club team to the Yandex Eats advertising unit: analytics and fast testing of ad hypotheses.
- Main Yandex Eats and Yandex Vendor webviews; micro-frontend architecture.
- An ad runner game whose first partner was Haval at its Russian launch; most of the banner ads, analytics debugging and new metrics.

VK, Delivery Club — Jun 2020 – Oct 2022
Lead Developer

- Maintained delivery-club.ru, effectively full-stack: admin panels for partners and lawyers, ad formats, internal UI and API libraries.
- After an incident, built a document tool for the legal team with versioning and audit logs.

Data-screen, Moscow — Jun 2019 – Mar 2020
Full-stack Developer

- A sports streaming service built from high-level ideas with full technical ownership (Node.js, React, PostgreSQL, Redis, Docker); COVID put it on hold.

Smart Solution, Moscow — Nov 2015 – Dec 2018
Senior Frontend Developer

- Frontend for about ten high-traffic sites in the Alexa top 50: legacy rewrites, optimisation and builds; ad integrations; trained developers and reviewed code.

Oversun, Web Developer — 2014 – 2015
WordPress sites from scratch with custom modules; led a junior developer on the company website.

Ideegroupe, Sochi, Web Developer — 2013
WordPress and ModX: modules for a city news portal, SQL optimisation.

RedBox, Buryatia, HTML Developer — 2011 – 2012
Joomla and WordPress templates, jQuery.

Skills

Generative models	Qwen Image, H3, Wan 2.2, LTX, SD, SDXL, TRELLIS.2, Kokoro, Higgs TTS, Whisper, Qwen3-VL
ML engineering	LoRA training, datasets, evaluation (identity similarity, LPIPS, lip-sync, held-out sets), CLIP probes, embeddings and RAG, INT8 and GGUF inference
Serving	ComfyUI, vLLM, ONNX Runtime, RunPod, GPU scheduling, Redis/BullMQ
LLM and agents	Tool use, MCP, Claude, GPT, Grok, Qwen, pgvector, Qdrant
Software	Python, TypeScript, Node.js, Bun, Hono, Next.js, React, PostgreSQL, ClickHouse, Kafka, Docker

Details

Languages	Russian (native), English (upper intermediate, B2)
Education	Secondary education
Work	Full-time, part-time or project work; remote, hybrid or on-site; open to business travel and relocation within Russia.